Clickbait Detector — Final Project

Presentation & Poster Content Guide · RMCS Binus University

PART 1 — PRESENTATION SLIDES

Slide 1: Title Slide

• Title: Enhancing YouTube Clickbait Detection Using Comment Sentiment Analysis and Title-Transcript Semantic Similarity

• All team member names + student IDs

• Course: RMCS — Binus University, Semester 4

• Presentation date

Slide 2: Background / Motivation

• YouTube has billions of videos — clickbait is a growing, widespread problem

• Users waste time on content that doesn't deliver what the title promises

• Existing detectors rely only on surface-level metadata (titles, view counts, likes)

• Two important signals are being ignored:

• — What viewers say in comments (negative sentiment = disappointment)

• — Whether the title actually matches the video content

Slide 3: Problem Statement

• How can we improve YouTube clickbait detection by incorporating:

• (1) Viewer comment sentiment, and

• (2) Semantic mismatch between title and transcript?

• Research gap: No prior work combines VADER comment sentiment + Sentence-BERT title-to-transcript similarity in a single pipeline

Slide 4: Related Work

• Vierti (2019) — SVM baseline model + open dataset (what this project builds on)

• Gothankar et al. (2022) — suggested title-transcript similarity as a clickbait signal

• Elyashar et al. (2022) — showed comment sentiment correlates with clickbait

• Bronakowski et al. (2023) — semantic features achieved 98% accuracy on news headlines

• Ahmadi & Chowanda (2023) — validated title-content matching for news clickbait

■ Keep to ~4-5 references. One sentence per reference is enough.

Slide 5: Proposed System Overview (Pipeline Diagram)

• Show a flowchart: Raw Dataset → Stage 1 (Extraction & Filtering) → Stage 2 (Feature Engineering: Block A, B, C) → Stage 3 (Training & Evaluation)

• Highlight the 3 feature blocks as the novel contribution

■ This should be a visual diagram, not a text slide. Use the flowchart from your report if available.

Slide 6: Dataset

- Source: Vierti's open YouTube dataset (~36,000 labeled videos)
- After filtering for available transcripts + comments: 9,393 videos
- After balancing 1:1 (clickbait : non-clickbait): 5,722 videos
- Train / Test split: 4,577 train (80%) / 1,145 test (20%) — stratified
- Data collection challenges:
- — YouTube API daily quota limits → required API key rotation
- — IP bans on transcript extraction → used cookie spoofing + exponential backoff

Slide 7: Feature Engineering — Block A (Baseline)

- Word2Vec title embeddings (trained on dataset titles) → 25 dimensions
- Params: vector_size=25, window=20, min_count=1, epochs=30
- Engagement metrics → 4 dimensions (log-scaled + min-max normalized):
- log(1+views), log(1+likes), log(1+dislikes), log(1+comments)
- Total Block A: 29 features
- This replicates Vierti's original feature set (Model 1 baseline)

Slide 8: Feature Engineering — Block B (Comment Sentiment)

- VADER Sentiment Analyzer applied to top 100 comments per video
- 4 features extracted:
- — mean_compound: average sentiment polarity
- — std_compound: standard deviation of sentiment
- — pct_positive: % comments with compound score > 0.3
- — pct_negative: % comments with compound score < -0.3
- Rationale: Angry or disappointed viewers leave negative comments under clickbait videos

Slide 9: Feature Engineering — Block C (Title-Transcript Similarity)

- Sentence-BERT (all-MiniLM-L6-v2) encodes both:
- — Video title → embedding vector
- — Video transcript → embedding vector
- Cosine similarity between title and transcript → 1 feature
- High similarity = title matches content (likely genuine)
- Low similarity = misleading title (likely clickbait)
- Note: Transcripts truncated at 256 subword tokens by the tokenizer

Slide 10: Models Compared

- Model 1 — Baseline SVM: Block A only (29-dim), Vierti's original hyperparams (C=3.7, γ=4.1)
- Model 2 — Enhanced SVM: Blocks A+B+C (34-dim), tuned via grid search (C=10, γ=scale)
- Model 3 — Enhanced Random Forest: Blocks A+B+C (34-dim), tuned via grid search
- Hyperparameter tuning: 5-fold cross-validation optimizing F1-score
- Feature dimensionality: 25 (W2V) + 4 (engagement) + 4 (VADER) + 1 (SBERT) = 34

■ Show this as a comparison table on the slide.

Slide 11: Results

- Test set: 1,145 samples (held-out, never seen during training)

Model	Accuracy	Precision	Recall	F1-Score

Model 1: Baseline SVM	95.81%	96.45%	95.10%	95.77%
Model 2: Enhanced SVM ★	**96.16%**	**96.32%**	**95.98%**	**96.15%**
Model 3: Enhanced RF	93.28%	94.12%	92.31%	93.20%

- ★ Model 2 best overall — +0.35% accuracy and +0.88% recall vs. baseline
- Model 2 false negatives: 23 (vs 28 in baseline) — 5 fewer missed clickbaits
- Also show confusion matrices for all 3 models on this slide

Slide 12: Discussion / Analysis

- Enhanced SVM improved accuracy (+0.35%) and recall (+0.88%) over baseline
- Comment sentiment and title-transcript gap are statistically valid signals ✓
- Random Forest underperformed: RBF-SVM handles dense continuous vectors better
- Trade-off: small accuracy gain requires significant extra data collection effort
- (API quota management, cookie spoofing, Sentence-BERT encoding overhead)

Slide 13: Demo Video

- Embed a short demo video here (Insert → Video in PowerPoint)
- Show: input a YouTube video URL → pipeline runs feature extraction → model outputs CLICKBAIT or NOT CLICKBAIT
- Ideal length: 1-2 minutes

■ This is its own dedicated slide. Just the video + a minimal title is enough.

Slide 14: Conclusion

- Successfully extended Vierti's SVM baseline with 2 new feature blocks (Block B + C)
- Best model: Enhanced SVM at 96.16% accuracy (F1: 96.15%)
- Comment sentiment (VADER) and title-transcript mismatch (Sentence-BERT) are effective clickbait signals
- Limitation: dataset reduced from ~36,000 to 5,722 due to transcript/comment availability
- Limitation: potential selection bias — only videos with public comments and transcripts

Slide 15: Future Work

- Scale up data collection with more API keys and faster extraction pipelines
- Add visual features — thumbnail analysis (color, face detection, text overlay)
- Try transformer-based classifiers (BERT, RoBERTa) instead of SVM
- Deploy as a real-time browser extension for YouTube
- Address selection bias by including videos without transcripts (e.g., audio transcription)

Slide 16: Q&A; / Thank You

- Team member names
- GitHub repository link (if public)
- Acknowledgement of advisor: Rhio Sutoyo

PART 2 — ACADEMIC POSTER STRUCTURE

Academic posters are typically A0 or A1 in portrait orientation. They are read in a Z or column pattern. Below is the recommended layout and content for each section.

HEADER (full width)
Title · Authors · Institution · Course

LEFT COLUMN (~33% width)	**MIDDLE COLUMN** (~33% width)	**RIGHT COLUMN** (~33% width)
• Background • Problem Statement • Related Work • Dataset	• Methodology / System Pipeline • Feature Engineering (Block A, B, C)	• Results (table + charts) • Discussion • Conclusion • Future Work / References

Header (Full Width)

- Title of the research (can be shortened from the paper title)
- All author names and student IDs
- Bina Nusantara University · RMCS · Semester 4 · Year
- Optional: University logo or department banner

1. Background & Problem Statement

- 2-3 sentences: why clickbait detection matters
- What existing approaches miss (no comment sentiment, no title-transcript check)
- Research question in 1 clear sentence

2. Related Work

- 3-4 key references in brief (1 line each)
- Vierti (2019), Gothankar et al. (2022), Elyashar et al. (2022), Bronakowski et al. (2023)
- What gap this work fills

3. Dataset

- Source: Vierti's dataset (~36,000 videos)
- After filtering: 5,722 balanced videos (2,861 clickbait / 2,861 non-clickbait)
- Train / Test: 4,577 / 1,145 — stratified 80/20 split
- Include a small data collection pipeline diagram

4. Methodology / System Pipeline

- A clean visual flowchart is essential here — this is the centrepiece of the poster
- Show: Dataset → Stage 1 (Extract & Filter) → Block A / Block B / Block C → Concatenate → SVM / RF → Output
- Label each block with the key technique (Word2Vec, VADER, Sentence-BERT)

5. Feature Engineering

- Block A: Word2Vec title embeddings (25-dim) + engagement metrics (4-dim) = 29 features
- Block B: VADER sentiment on top-100 comments → 4 features (mean, std, %pos, %neg)
- Block C: Sentence-BERT cosine similarity (title ↔ transcript) → 1 feature
- Total enhanced feature vector: 34 dimensions
- Use icons or small diagrams — avoid paragraph text

6. Results

- Comparison table (Model 1 vs 2 vs 3): Accuracy, Precision, Recall, F1
- Bar chart or grouped bar chart is more visually impactful than a table alone
- Highlight the best result (Model 2: 96.16% accuracy, 96.15% F1)
- Include at least one confusion matrix (preferably Model 2)
- Key callout stat: '+0.35% accuracy and 5 fewer missed clickbaits'

7. Discussion

- Why Model 2 beat baseline: comment sentiment + semantic gap are valid signals
- Why Random Forest underperformed: RBF-SVM handles dense continuous vectors better
- Limitation: dataset shrunk significantly due to transcript/comment availability
- 1-2 bullet points maximum — keep it tight on a poster

8. Conclusion

- Best model: Enhanced SVM, 96.16% accuracy
- Comment sentiment (VADER) and title-transcript mismatch (SBERT) are effective features
- Open-source reproducible pipeline
- 2-3 bullets only

9. Future Work & References

- Future: thumbnail analysis, transformer classifiers (BERT/RoBERTa), browser extension
- References: 4-5 key citations in compact format (IEEE or APA)
- QR code linking to GitHub repo (optional but looks great on a poster)

POSTER DESIGN TIPS

- Size: A0 (84 × 119 cm) or A1 (59 × 84 cm) portrait is standard
- Font size: Title ≥ 72pt · Section headers ≥ 36pt · Body text ≥ 24pt (readable from 1 metre away)
- 60% of the poster should be visuals (diagrams, charts, tables) — not text
- Use a consistent 2 or 3 colour palette — stick to it throughout
- Every section should have at least one visual element (no text-only blocks)
- Keep body text to bullet points — remove full sentences wherever possible
- Leave white space — a crowded poster is harder to read than a sparse one
- The pipeline/system diagram is the most important visual — make it large and central

• QR code in the bottom corner linking to your GitHub or paper is a nice touch